

KNUST RANKS NO.1 GLOBALLY FOR THE
PROVISION OF QUALITY EDUCATION (SDG 4)



Continuous-time Signals and Systems

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Test 1

1. What is DSP?
2. State 5 advantages and 3 disadvantages of DSP over ASP.
3. Using Diagrams explain the DSP solution.
4. State any two applications of DSP in the following areas: Communications, Satellite Navigation, Measurement and Control, Military and Safety, Consumer Electronics, Medical



Introduction

Any problems about signal analyses and processing may be thought of as letting signals through systems.



- ❖ From $f(t)$ and $h(t)$, find $y(t)$, Signal processing
- ❖ From $f(t)$ and $y(t)$, find $h(t)$, System design
- ❖ From $y(t)$ and $h(t)$, find $f(t)$, Signal reconstruction



Introduction

- There are so many different signals and systems such that it is impossible to describe them one by one
- The best approach is to represent the signal as **a combination of some kind of simplest signals** which then passes through the system and produces a response. We then combine the responses of all the simple signals, which then becomes the system response for the original input signal.
- This is the basic approach to study the signal analyses and processing.



Continuous-time Signal

- ❖ All signals are thought of as a pattern of variations in time and represented as a time function $f(t)$.
- ❖ In the real-world, all signals have a start. Let the start be $t = 0$ that means

$$f(t) = 0 \quad t < 0$$

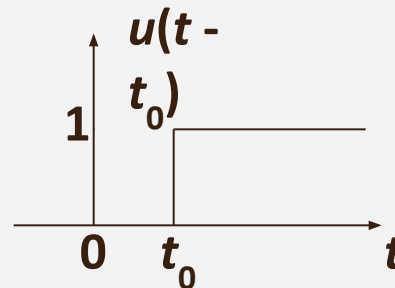
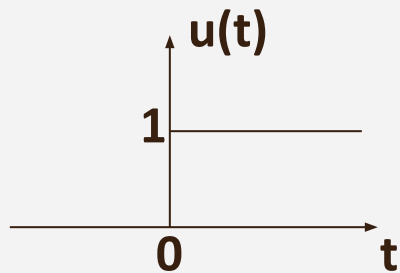
Such signals are called **causal signals**.



Typical signals and their representation

◆ Unit Step $u(t)$

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$



◆ Note, $u(t)$ is a basic causal signal, **multiply it with any non-causal signal** to get a causal signal.



Typical signals and their representation

Sinusoidal $A\sin(\omega t + \phi)$

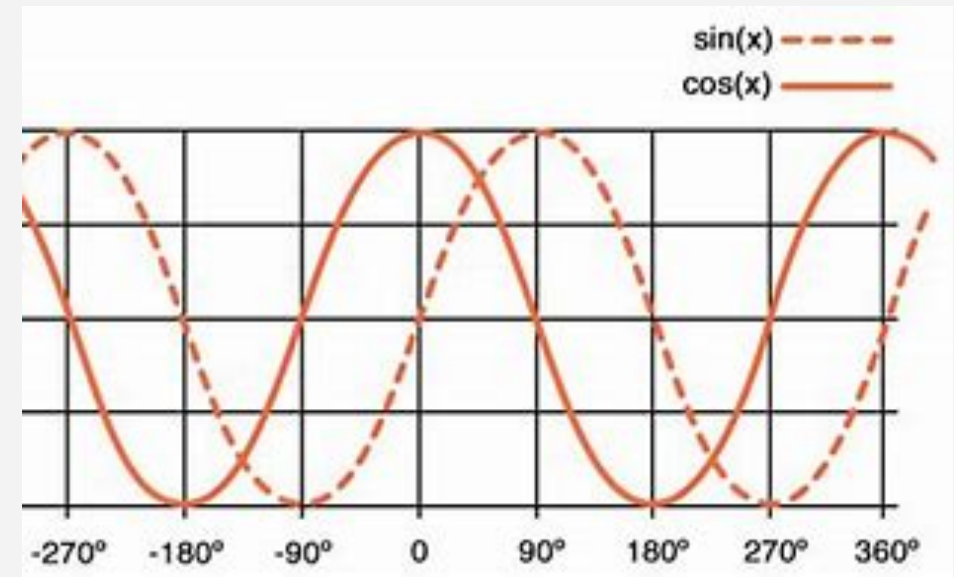
$$f(t) = A\sin(\omega t + \phi) = A\sin(2\pi f t + \phi)$$

A - Amplitude

f – frequency (Hz, cycle per second)

$\omega = 2\pi f$ angular frequency (radians/sec)

ϕ – start phase (radians)



Typical signals and their representation

◆ sin/cos signals may be represented by complex exponential

$$A \sin(\omega t + \varphi) = \frac{A}{2j} (e^{j(\omega t + \varphi)} - e^{-j(\omega t + \varphi)})$$

$$A \cos(\omega t + \varphi) = \frac{A}{2} (e^{j(\omega t + \varphi)} + e^{-j(\omega t + \varphi)})$$

◆ Euler's relation

$$e^{j(\omega t + \varphi)} = \cos(\omega t + \varphi) + j \sin(\omega t + \varphi)$$



Typical signals and their representation

- ❖ The sinusoidal signal is a basic *periodic* signal which is important both in theory and engineering.
- ❖ The sinusoidal signal is a non-causal signal. All periodic signals are non-causal because they have no start and no end.

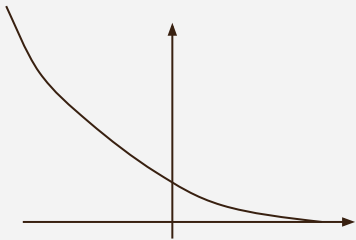
$$f(t) = f(t + mT), \quad m = \pm 1, \pm 2, \dots, \pm \infty$$



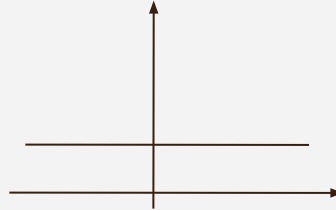
Typical signals and their representation

◆ Exponential $f(t) = e^{\alpha t}$

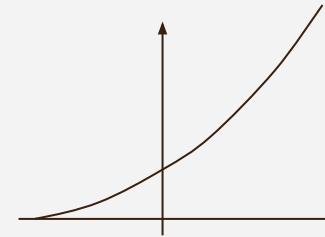
- α is real



$\alpha < 0$ decaying



$\alpha = 0$ constant



$\alpha > 0$ growing



Typical signals and their representation

◆ Exponential $f(t) = e^{\alpha t}$

- α is complex $\alpha = \sigma + j\omega$

$$\begin{aligned} f(t) &= Ae^{\alpha t} = Ae^{(\sigma + j\omega)t} \\ &= Ae^{\sigma t} \cos \omega t + j Ae^{\sigma t} \sin \omega t \end{aligned}$$

$\sigma = 0$, sinusoidal

$\sigma > 0$, growing sinusoidal

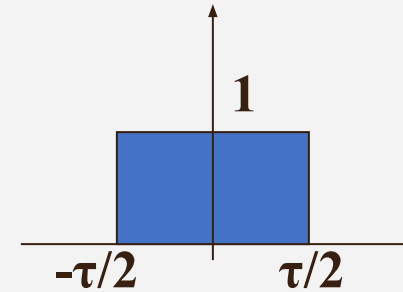
$\sigma < 0$, decaying sinusoidal (damped)



Typical signals and their representation

◆ Gate signal

$$p_{\tau}(t) = \begin{cases} 1 & |t| < \frac{\tau}{2} \\ 0 & |t| > \frac{\tau}{2} \end{cases}$$



The gate signal can be represented by unit step signals:

$$p_{\tau}(t) = u(t + \tau/2) - u(t - \tau/2)$$



Typical signals and their representation

◆ Unit Impulse Signal

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

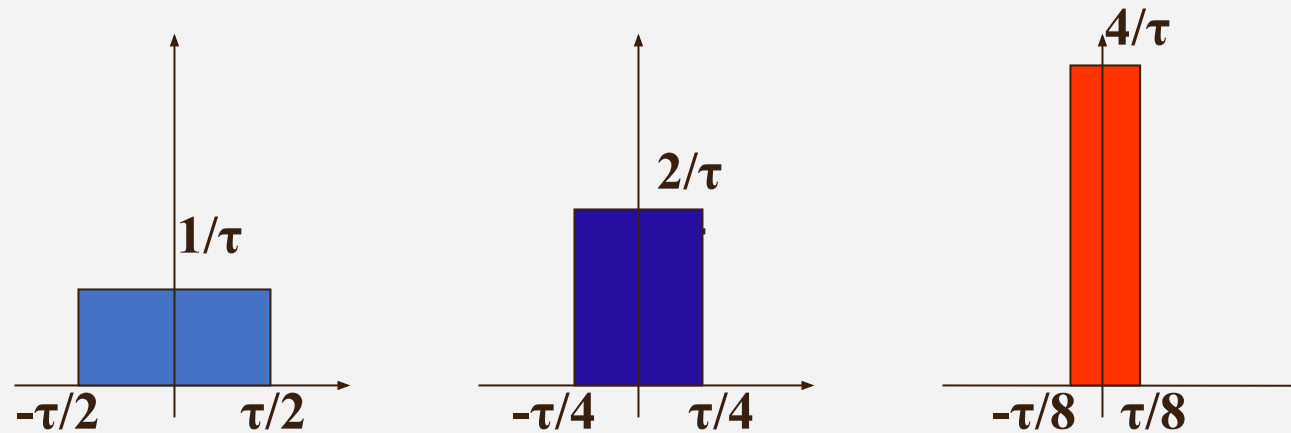
$$\delta(t) = 0 \quad t \neq 0$$

- $\delta(t)$ is non-zero only at $t = 0$, otherwise is 0
- $\delta(t)$ could not be represented by a constant even at $t = 0$, but by an integral.
- Regular functions have exact values at exact times. Obviously, $\delta(t)$ is not a Regular function, but a *singularity function* (or *generalized function*).



Unit impulse function $\delta(t)$

❖ With a gate signal $p_\tau(t)$, short the duration τ and keep the unit area



When $\tau \rightarrow 0$, the amplitude tends to ∞ , which means it is impossible to define $\delta(t)$ by a regular function.



Properties of $\delta(t)$

❖ Sampling Property

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = f(0)$$

Brief explanation :

- When $t \neq 0$, $\delta(t) = 0$, then $f(t) \cdot \delta(t) = 0$
- When $t = 0$, $f(t) = f(0)$ is a constant. Based on the definition of $\delta(t)$, it is easy to see that:

$$\int_{-\infty}^{\infty} f(t)\delta(t)dt = \int_{-\infty}^{\infty} f(0)\delta(t)dt = f(0) \int_{-\infty}^{\infty} \delta(t)dt = f(0)$$



Properties of $\delta(t)$

◆ $\delta(t)$ shift



◆ $\delta(t)$ times a constant A :

$$A\delta(t)$$

A is called impulse intension which is the area of the integral



Properties of $\delta(t)$

◆ Differential of $\delta(t)$ is also an impulse

$$\delta'(t) = d\delta(t)/dt \quad \text{and}$$
$$\int_{-\infty}^{\infty} f(t)\delta'(t)dt = -f'(0)$$

◆ Integral of $\delta(t)$ is unit step signal:

$$\int_{-\infty}^{\infty} \delta(t)dt = \int_{-\infty}^{0_-} \delta(t)dt + \int_{0_-}^{\infty} \delta(t)dt = u(t)$$



Properties of $\delta(t)$

◆ $\delta(t)$ is an even function, that is

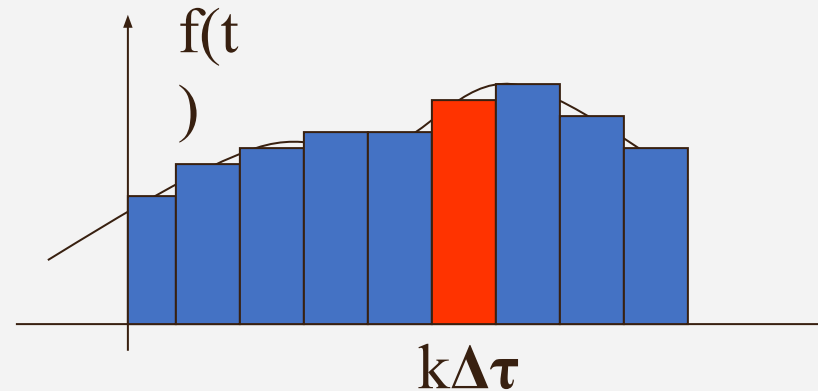
$$\delta(t) = \delta(-t)$$

We got $\delta(t)$ from a gate signal, and the gate signal is an even function. It is also easy to show the even property of $\delta(t)$.



Signal representation based on $\delta(t)$

❖ Any signal can be represented by a sum of shifted and weighted $\delta(t)$



Approximate $f(t)$ with a series of rectangles. For the rectangle $t=k\Delta\tau$, the duration is $\Delta\tau$, the height is $f(k\Delta\tau)$, and the area is $f(k\Delta\tau) \Delta\tau$. So the rectangle can be represented by $f(k\Delta\tau) \Delta\tau \delta(t - k\Delta\tau)$.



Signal representation based on $\delta(t)$

$f(t)$ can be approximated as follows:

$$f(t) \approx \sum_{k=-\infty}^{\infty} f(k\Delta\tau)\Delta\tau\delta(t - k\Delta\tau)$$

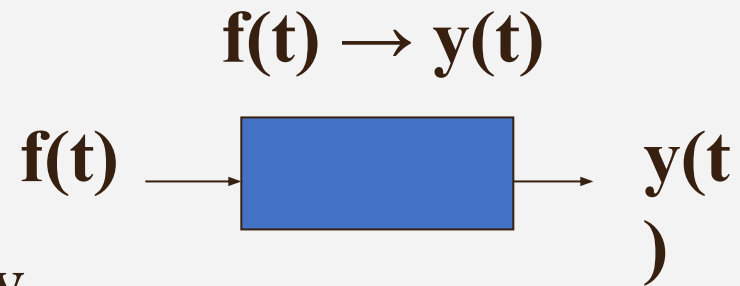
The smaller $\Delta\tau$, the higher the accuracy, and when $\Delta\tau \rightarrow d\tau$, $k\Delta\tau \rightarrow \tau$, the above expression becomes a precise representation:

$$f(t) = \int_{-\infty}^{\infty} f(\tau)\delta(t - \tau)d\tau$$



Linear time-invariant system

Input signal $f(t)$ into system, output $y(t)$:



If satisfy

$$af(t) \rightarrow ay(t) \quad \text{and} \quad f_1(t) + f_2(t) \rightarrow y_1(t) + y_2(t)$$

Scaling

additivity

That is $af_1(t) + bf_2(t) \rightarrow ay_1(t) + by_2(t)$

The system is called **linear**.



Linear time-invariant system

A system is time invariant If:

$$f(t - t_0) \rightarrow y(t - t_0)$$

The system is called **time-invariant**.

The reason why linear time-invariant systems are important is that based on $\delta(t)$ shifted and weighted sum representation of $f(t)$, we can look at the system response for a unit impulse, then make the sum of responses for all shifted and weighted impulses to get the whole response. This is possible only with linear time-invariant systems.



System unit impulse response

The system response for unit impulse $\delta(t)$ is called **system unit impulse response**, and denoted as $h(t)$



$h(t)$ is an important figure of linear time-invariant systems. It can be used to get the response for any input signal based on $h(t)$, because any signal can be represented as a sum of shifted and weighted $\delta(t)$.



When signals pass through linear time-invariant system

$$\delta(t) \rightarrow h(t)$$

$$a\delta(t - t_0) \rightarrow ah(t - t_0)$$

$$f(t) = \int_{-\infty}^{\infty} f(\tau)\delta(t - \tau)d\tau$$

$$y(t) = \int_{-\infty}^{\infty} f(\tau)h(t - \tau)d\tau$$

Denoted $y(t) = f(t) * h(t)$, which is called **convolution** of $f(t)$ and $h(t)$



Analyses in time domain and frequency domain

- With the time domain analysis, signals are decomposed as the sum of shifted and weighted $\delta(t)$, and only the system response for $\delta(t)$ has attracted attention.
- ❖ With the frequency domain analysis, signals are decomposed as the sum of fundamental $\sin\omega t$ and harmonics.



Fourier series of periodic signals

- Periodic signals can be expanded with Fourier series

$$f(t) = \sum_{n=0}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

$$a_0 = \frac{1}{T} \int_{t_1}^{T+t_1} f(t) dt \quad \text{The average in one period}$$

$$a_n = \frac{2}{T} \int_{t_1}^{T+t_1} f(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T} \int_{t_1}^{T+t_1} f(t) \sin n\omega_0 t dt$$



Fourier series of periodic signals

- Fourier series in exponential form

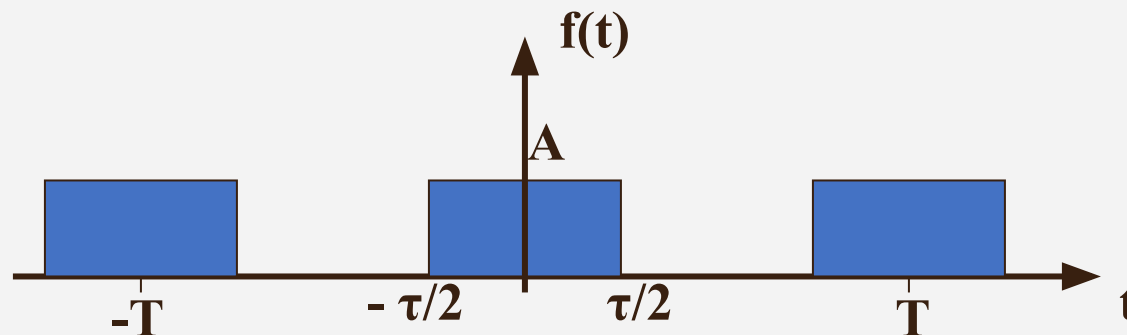
$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{jn\omega_0 t}$$

$$F_n = \frac{1}{T} \int_{t_1}^{T+t_1} f(t) e^{-jn\omega_0 t} dt$$



Spectrum of periodic signals

- Periodic rectangles with period T , duration τ , and amplitude A :

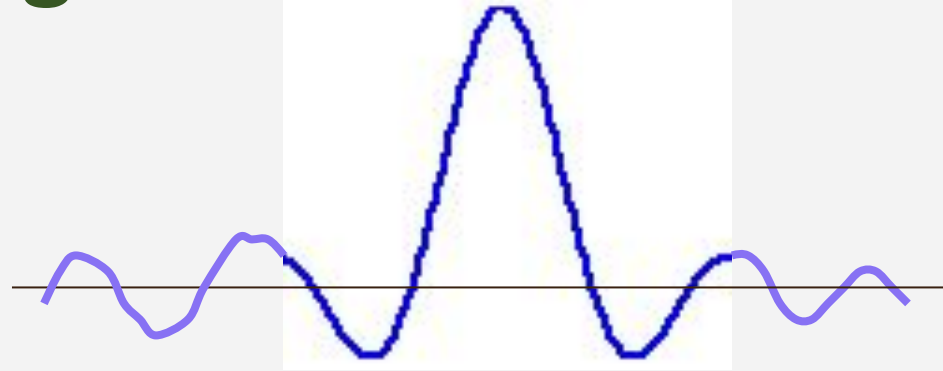


Periodic signals is expanded to Fourier series

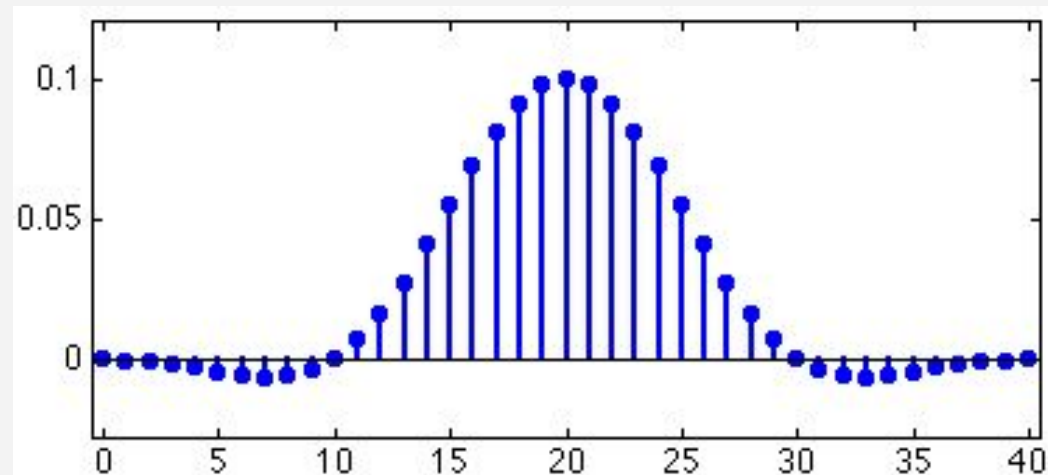
$$\begin{aligned} F_n &= \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \\ &= \frac{1}{T} \int_{-\tau/2}^{\tau/2} A e^{-jn\omega_0 t} dt \\ &= \frac{2A \sin \frac{n\omega_0 \tau}{2}}{n\omega_0 T} \\ &= \frac{A\tau}{T} \frac{\sin x}{x} \quad x = \frac{n\omega_0 \tau}{2} \end{aligned}$$



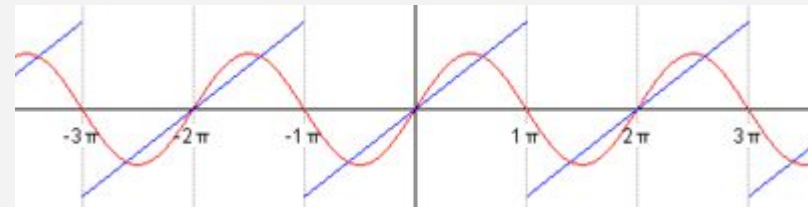
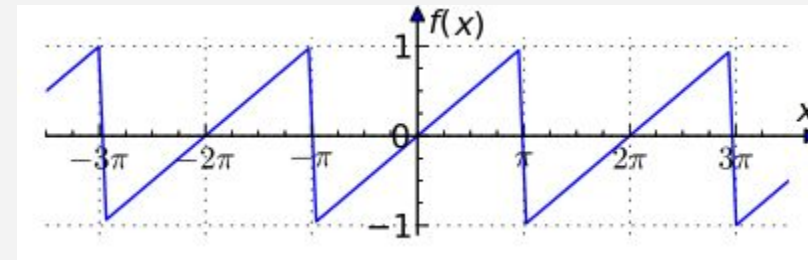
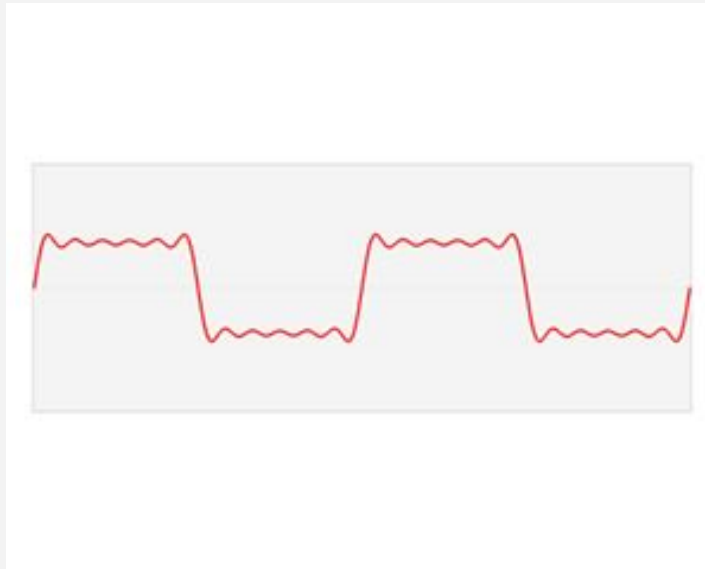
Sampling function $\text{Sa}(x) = \sin x / x$



$$F_n = (A\tau/T)\text{Sa}(n\omega_0 \tau / 2)$$



Fourier series of periodic signals



Fourier analyses of non-periodic signals—Fourier Transform

When the period of a periodic signal is expanded to ∞ , the signal becomes non-periodic:

$$F_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \quad \omega_0 = 2\pi/T$$

But $T \rightarrow \infty$, F_n and $\omega_0 \rightarrow 0$, the amplitude of lines and the distance between lines tend to 0. It is unreasonable to describe the signal with frequency lines. So the concept of **spectrum density** is introduced.



Fourier Transform pair

$$F(j\omega) = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

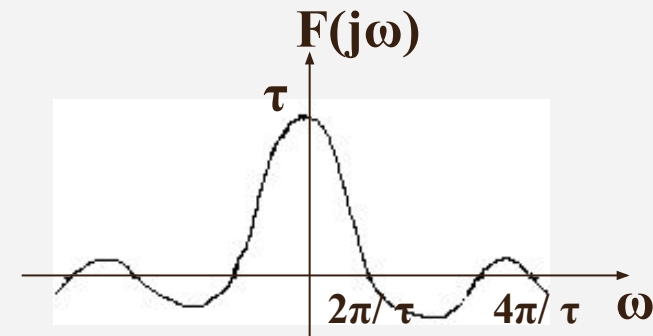
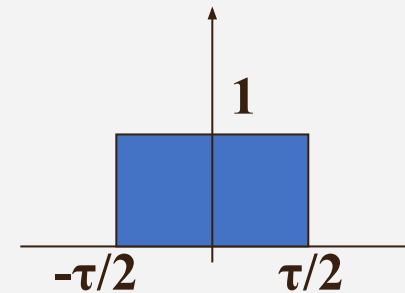
$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega)e^{j\omega t} d\omega$$



Fourier Transform of typical signals

- Gate

$$\begin{aligned} F(j\omega) &= \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \\ &= 2 \int_0^{\infty} p_{\tau}(t) \cos \omega t dt \\ &= 2 \int_0^{\tau/2} \cos \omega t dt \\ &= \frac{2}{\omega} \sin \frac{\omega \tau}{2} \\ &= \tau \text{Sa}\left(\frac{\omega \tau}{2}\right) \end{aligned}$$

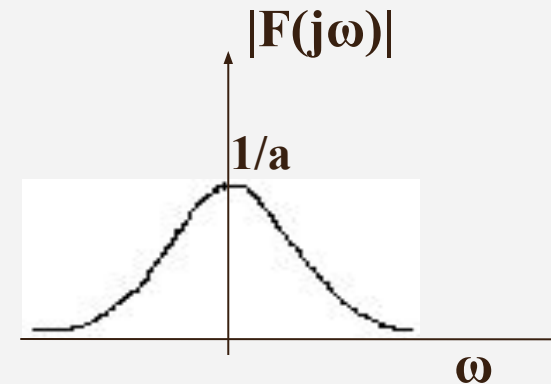
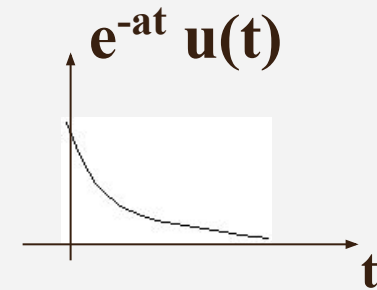


Fourier Transform of typical signals

- Exponential $f(t) = e^{-at} u(t) \quad a > 0$

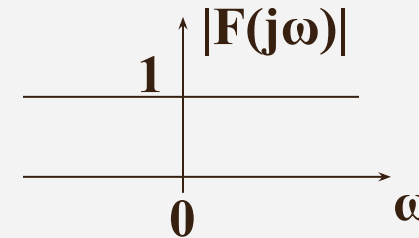
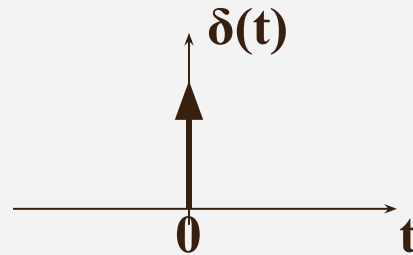
$$\begin{aligned} F(j\omega) &= \int_0^{\infty} e^{-at} e^{-j\omega t} dt \\ &= \frac{1}{a + j\omega} \end{aligned}$$

$$\begin{aligned} |F(j\omega)| &= \frac{1}{\sqrt{a^2 + \omega^2}} \\ \varphi(\omega) &= -\tan^{-1} \frac{\omega}{a} \end{aligned}$$



Fourier Transform of typical signals

- Unit impulse $\delta(t)$



$$F[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

Unit impulse has uniform frequency density in whole frequency range, that means it has infinite wide band.



Fourier Transform of typical signals

- Constant 1

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

This result could be obtained directly based on the symmetry of Fourier Transform.

Constant 1 represents direct current signal, and its spectrum is non-zero only at $\omega = 0$, which is at $\delta(\omega)$



Fourier Transform of typical signals

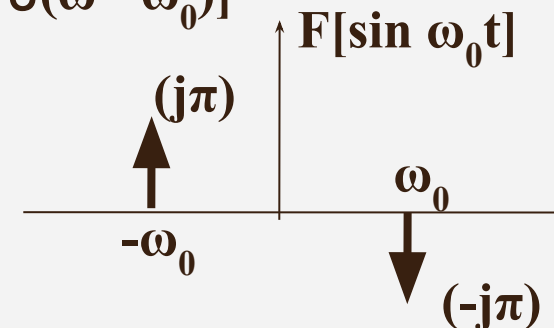
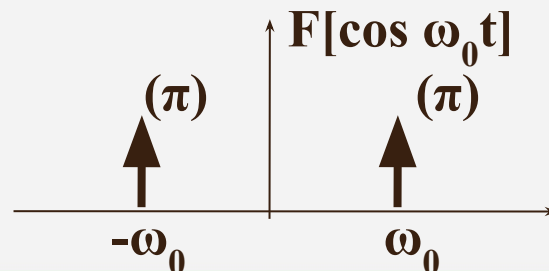
▪ Sin and cos function

Based on the transform pair $1 \longleftrightarrow 2\pi\delta(\omega)$ and $\delta(t) \longleftrightarrow 1$, we have some important conclusions:

$$F[e^{j\omega_0 t}] = \int_0^\infty e^{j\omega_0 t} e^{-j\omega t} dt = \int_0^\infty e^{-j(\omega - \omega_0)t} dt = 2\pi\delta(\omega - \omega_0)$$

$$F[\cos\omega_0 t] = F[(e^{j\omega_0 t} + e^{-j\omega_0 t})/2] = \pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$$

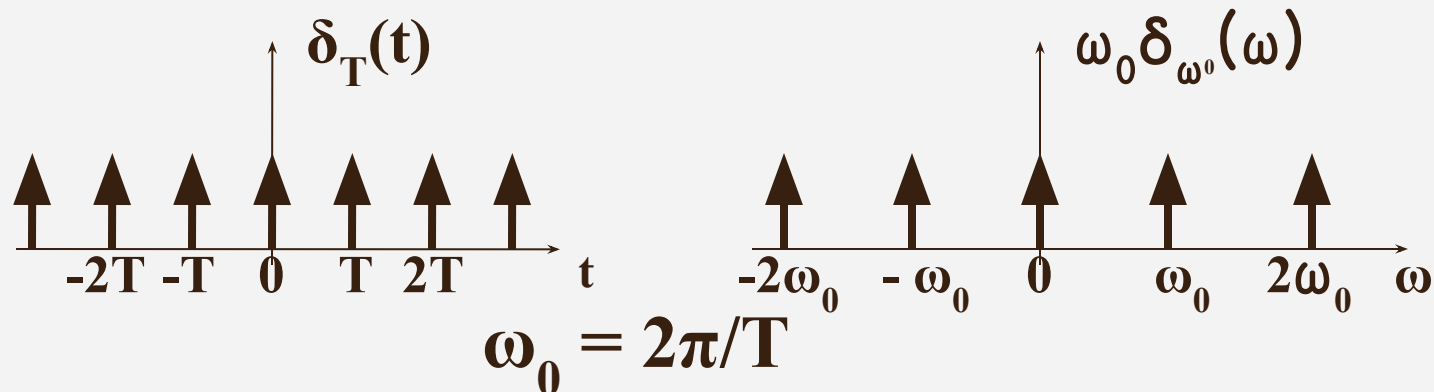
$$F[\sin \omega_0 t] = F[(e^{j\omega_0 t} - e^{-j\omega_0 t})/2j] = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$$



Fourier Transform of typical signals

- Unit impulse sequence

$$\delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$
$$F[\delta_T(t)] = \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$
$$= \omega_0 \delta_{\omega_0}(\omega)$$



Properties of Fourier Transform

- Linear

Fourier Transform is an integral, and it is a linear operation:

$$\text{If } f_1(t) \longleftrightarrow F_1(j\omega) \quad f_2(t) \longleftrightarrow F_2(j\omega)$$

$$\text{Then } af_1(t) + bf_2(t) \longleftrightarrow aF_1(j\omega) + bF_2(j\omega)$$



Properties of Fourier Transform

- Time shift

Signal's shift in time domain equals phase shift in frequency domain

$$f(t - t_0) \longleftrightarrow F(j\omega)e^{-j\omega t_0}$$

Based on the definition of Fourier Transform, the above result is easy to be shown.



Properties of Fourier Transform

- Frequency shift (Modulation theorem)

Modulate carrier $\sin\omega_0 t$ with base band signal $f(t)$

Based on definition of Fourier Transform:

$$f(t)e^{j\omega_0 t} \longleftrightarrow F[j(\omega - \omega_0)]$$

With Euler's relation, it is easy to show:

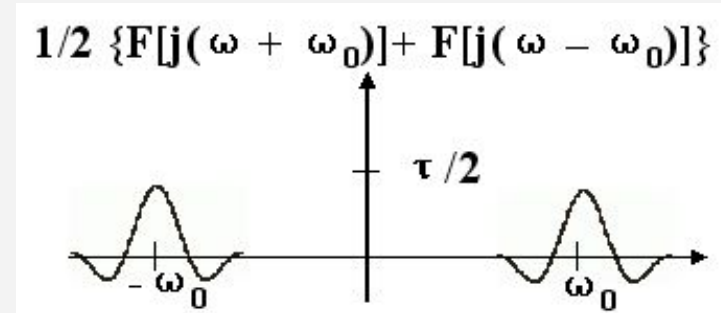
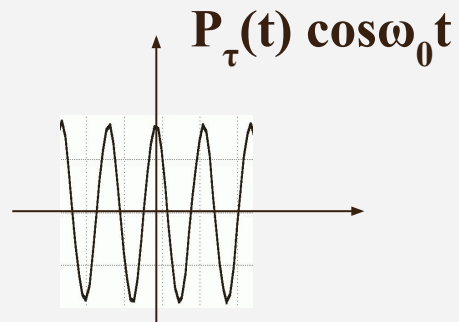
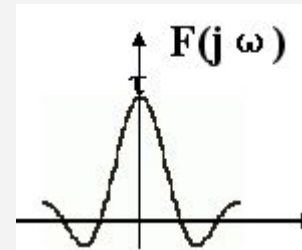
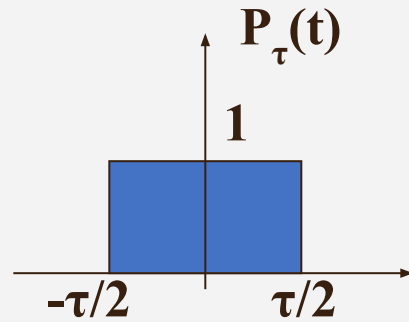
$$f(t) \cos\omega_0 t = 1/2 f(t) (e^{j\omega_0 t} + e^{-j\omega_0 t})$$

$$f(t) \cos\omega_0 t \longleftrightarrow 1/2 \{F[j(\omega + \omega_0)] + F[j(\omega - \omega_0)]\}$$

$$f(t) \sin\omega_0 t \longleftrightarrow j/2 \{F[j(\omega + \omega_0)] - F[j(\omega - \omega_0)]\}$$



Spectrum of amplitude modulation



Properties of Fourier Transform

- Energy theorem

$$W = \int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(j\omega)|^2 d\omega$$

It is called **Parseval's relation**.

W is energy of the signal, $|F(j\omega)|^2$ named **signal energy spectrum** is the signal energy in unit frequency band and has a similar shape with $|F(j\omega)|$, but no phase information.



Properties of Fourier Transform

- Convolution theorem

- **convolution theorem in time domain**

$$f_1(t) * f_2(t) \longleftrightarrow F_1(j\omega) F_2(j\omega)$$

- **convolution theorem in frequency domain**

$$f_1(t) f_2(t) \longleftrightarrow 1/(2\pi)[F_1(j\omega) * F_2(j\omega)]$$

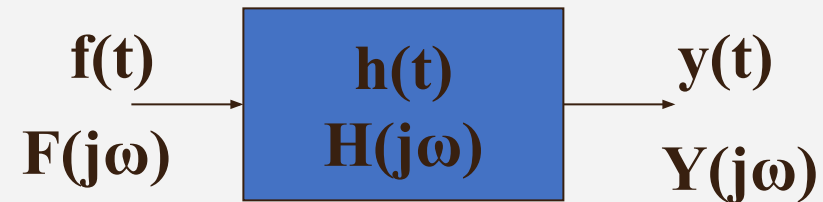


Properties of Fourier Transform

- Transfer the convolution operation in one domain to the algebra operation in another domain. Almost all of the properties discussed above can be shown based on the convolution theorem.



Fourier analyses of linear system



$$y(t) = f(t) * h(t)$$

$$Y(j\omega) = F(j\omega) H(j\omega)$$

- $H(j\omega)$ is called **system function**, or **transfer function**



Fourier analyses of linear system

- Spectrum analyses is an active research area today. If the system is too complicated to be represented by analytical expressions, the solution is to use input sine signals with different frequencies and measure the system output, then combine them to give the system transfer function.
- This is called a **frequency sweep**



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